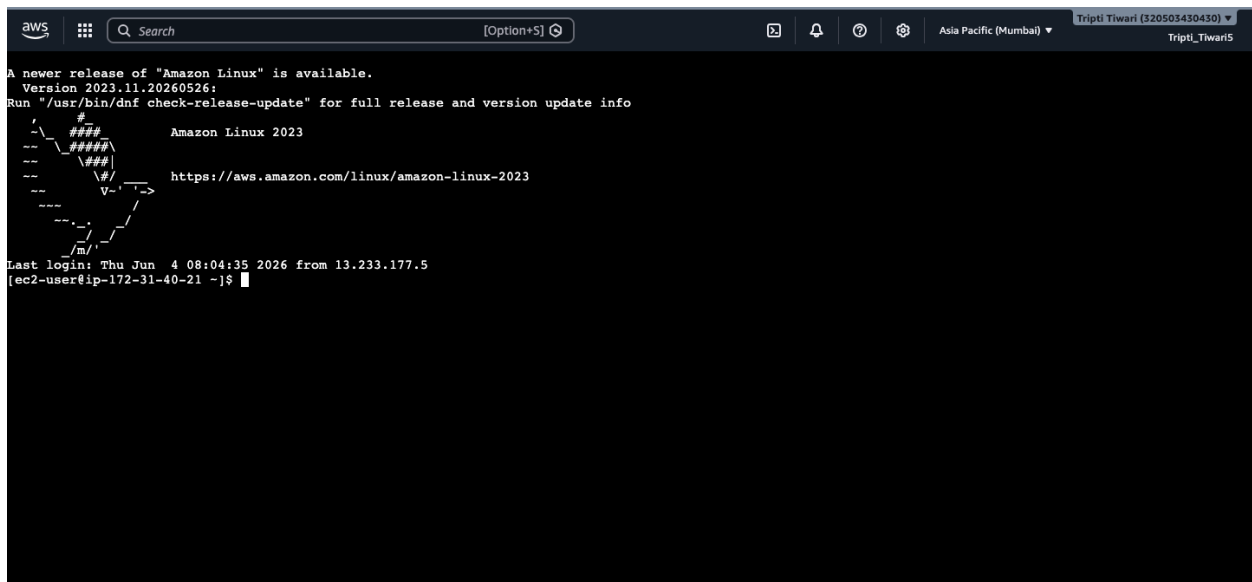


Docker Training Guide with Windows and EC2 Setup

Docker Installation on AWS EC2 (Amazon Linux)

STEP 1:- Connect to EC2

`ssh -i key.pem ec2-user@<public-ip>`

A screenshot of an AWS terminal window. The terminal shows the login process for an Amazon Linux EC2 instance. At the top, there's a notification about a newer release of Amazon Linux being available. Below that, the terminal displays the Amazon Linux logo and version information. The user is logged in as 'ec2-user' on the instance 'ip-172-31-40-21'. The terminal prompt is '[ec2-user@ip-172-31-40-21 ~]\$'.

STEP 2:- Install Docker

`sudo yum update -y`

```
~/m/
Last login: Thu Jun  4 08:04:35 2026 from 13.233.177.5
[ec2-user@ip-172-31-40-21 ~]$ sudo yum update -y
Last metadata expiration check: 11:43:04 ago on Thu Jun  4 06:59:29 2026.
=====
WARNING:
  A newer release of "Amazon Linux" is available.

  Available Versions:

    Version 2023.11.20260526:
      Run the following command to upgrade to 2023.11.20260526:

        dnf upgrade --releasever=2023.11.20260526

  Release notes:
    https://docs.aws.amazon.com/linux/al2023/release-notes/relnotes-2023.11.20260526.html
=====
Dependencies resolved.
Nothing to do.
Complete!
[ec2-user@ip-172-31-40-21 ~]$
```

sudo yum install docker -y

```
sudo: hdiutil: command not found
[ec2-user@ip-172-31-40-231 ~]$ ssh -i mynewkey.pem ec2-user@13.201.26.131
Warning: Identity file mynewkey.pem not accessible: No such file or directory.
The authenticity of host '13.201.26.131 (13.201.26.131)' can't be established.
ED25519 key fingerprint is SHA256:zGKa9w5Xseffe+KuNTizoXiJ4sEy2AoAacGCJqjALLU.
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '13.201.26.131' (ED25519) to the list of known hosts.
ec2-user@13.201.26.131: Permission denied (publickey,gssapi-keyex,gssapi-with-mic).
[ec2-user@ip-172-31-40-231 ~]$ sudo yum update -y
Amazon Linux 2023 Kernel Livepatch repository                                     389 kB/s | 47 kB    00:00
Dependencies resolved.
Nothing to do.
Complete!
[ec2-user@ip-172-31-40-231 ~]$ sudo yum install docker -y
Last metadata expiration check: 0:00:12 ago on Fri Jun  5 07:25:41 2026.
Dependencies resolved.
=====
Package                                Architecture      Version                                Repository          Size
=====
Installing:
docker                                x86_64            25.0.14-1.amzn2023.0.6               amazonlinux          46 M
Installing dependencies:
container-selinux                     noarch            4:2.245.0-1.amzn2023                 amazonlinux          58 k
containerd                             x86_64            2.2.3-1.amzn2023.0.2                 amazonlinux          24 M
=====
```

i-0b9340441837baa29 (my new server)

PublicIPs: 13.201.26.131 PrivateIPs: 172.31.40.231

CloudShell Feedback Console Mobile App

© 2026, Amazon Web Services, Inc. or its affiliates. Privacy Terms Cookie preferences

aws

Search

[Option+S]

Asia Pacific (Mumbai) Tripti Tiwari (s20503430430) Tripti_Tiwari5

Package	Architecture	Version	Repository	Size
Installing:				
docker	x86_64	25.0.14-1.amzn2023.0.6	amazonlinux	46 M
Installing dependencies:				
container-selinux	noarch	4:2.245.0-1.amzn2023	amazonlinux	58 k
containerd	x86_64	2.2.3-1.amzn2023.0.2	amazonlinux	24 M
iptables-libs	x86_64	1.8.8-3.amzn2023.0.2	amazonlinux	401 k
iptables-nft	x86_64	1.8.8-3.amzn2023.0.2	amazonlinux	183 k
libcgroup	x86_64	3.0-1.amzn2023.0.1	amazonlinux	75 k
libnetfilter_conntrack	x86_64	1.0.8-2.amzn2023.0.2	amazonlinux	58 k
libnftnl	x86_64	1.0.1-19.amzn2023.0.2	amazonlinux	30 k
libnftnl	x86_64	1.2.2-2.amzn2023.0.2	amazonlinux	84 k
pigz	x86_64	2.5-1.amzn2023.0.3	amazonlinux	83 k
runC	x86_64	1.3.4-5.amzn2023.0.2	amazonlinux	3.9 M

Transaction Summary

Install 11 Packages

Total download size: 76 M
Installed size: 284 M
Downloading Packages:

(1/11): container-selinux-2.245.0-1.amzn2023.noarch.rpm	1.4 MB/s	58 kB	00:00
(2/11): iptables-libs-1.8.8-3.amzn2023.0.2.x86_64.rpm	13 MB/s	401 kB	00:00
(3/11): iptables-nft-1.8.8-3.amzn2023.0.2.x86_64.rpm	7.6 MB/s	183 kB	00:00
(4/11): libcgroup-3.0-1.amzn2023.0.1.x86_64.rpm	2.5 MB/s	75 kB	00:00
(5/11): libnetfilter_conntrack-1.0.8-2.amzn2023.0.2.x86_64.rpm	2.0 MB/s	58 kB	00:00
(6/11): libnftnl-1.0.1-19.amzn2023.0.2.x86_64.rpm	1.0 MB/s	30 kB	00:00
(7/11): libnftnl-1.2.2-2.amzn2023.0.2.x86_64.rpm	2.8 MB/s	84 kB	00:00
(8/11): pigz-2.5-1.amzn2023.0.3.x86_64.rpm	2.9 MB/s	83 kB	00:00
(9/11): containerd-2.2.3-1.amzn2023.0.2.x86_64.rpm	66 MB/s	24 MB	00:00
(10/11): runC-1.3.4-5.amzn2023.0.2.x86_64.rpm	24 MB/s	3.9 MB	00:00
(11/11): docker-25.0.14-1.amzn2023.0.6.x86_64.rpm	68 MB/s	46 MB	00:00

i-0b9340441837baa29 (my new server)

PublicIPs: 13.201.26.131 PrivateIPs: 172.31.40.231

CloudShell Feedback Console Mobile App

© 2026, Amazon Web Services, Inc. or its affiliates. Privacy Terms Cookie preferences

```
aws [Option+S] Asia Pacific (Mumbai) Tripti Tiwari (320503430430) Tripti_Tiwari5
(9/11): containerd-2.2.3-1.amzn2023.0.2.x86_64.rpm 66 MB/s 24 MB 00:00
(10/11): runc-1.3.4-5.amzn2023.0.2.x86_64.rpm 24 MB/s 3.9 MB 00:00
(11/11): docker-25.0.14-1.amzn2023.0.6.x86_64.rpm 68 MB/s 46 MB 00:00
-----
Total 105 MB/s | 76 MB 00:00
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing :
  Installing : runc-1.3.4-5.amzn2023.0.2.x86_64 1/11
  Installing : containerd-2.2.3-1.amzn2023.0.2.x86_64 2/11
  Running scriptlet: containerd-2.2.3-1.amzn2023.0.2.x86_64 2/11
  Installing : pigz-2.5-1.amzn2023.0.3.x86_64 3/11
  Installing : libnftnl-1.2.2-2.amzn2023.0.2.x86_64 4/11
  Installing : libnftnl-1.0.1-19.amzn2023.0.2.x86_64 5/11
  Installing : libnftfilter_conntrack-1.0.8-2.amzn2023.0.2.x86_64 6/11
  Installing : iptables-libs-1.8.8-3.amzn2023.0.2.x86_64 7/11
  Installing : iptables-nft-1.8.8-3.amzn2023.0.2.x86_64 8/11
  Running scriptlet: iptables-nft-1.8.8-3.amzn2023.0.2.x86_64 8/11
  Installing : libgroup-3.0-1.amzn2023.0.1.x86_64 9/11
  Running scriptlet: container-selinux-4:2.245.0-1.amzn2023.noarch 10/11
  Installing : container-selinux-4:2.245.0-1.amzn2023.noarch 10/11
  Running scriptlet: container-selinux-4:2.245.0-1.amzn2023.noarch 10/11
  Running scriptlet: docker-25.0.14-1.amzn2023.0.6.x86_64 11/11
  Installing : docker-25.0.14-1.amzn2023.0.6.x86_64 11/11
  Running scriptlet: docker-25.0.14-1.amzn2023.0.6.x86_64 11/11
Created symlink /etc/systemd/system/sockets.target.wants/docker.socket → /usr/lib/systemd/system/docker.socket.

Running scriptlet: container-selinux-4:2.245.0-1.amzn2023.noarch 11/11
Running scriptlet: docker-25.0.14-1.amzn2023.0.6.x86_64 11/11
Verifying : container-selinux-4:2.245.0-1.amzn2023.noarch 1/11
Verifying : containerd-2.2.3-1.amzn2023.0.2.x86_64 2/11
Verifying : docker-25.0.14-1.amzn2023.0.6.x86_64 3/11
Verifying : iptables-libs-1.8.8-3.amzn2023.0.2.x86_64 4/11
```

i-0b9340441837baa29 (my new server)

PublicIPs: 13.201.26.131 PrivateIPs: 172.31.40.231

```
aws [Option+S] Asia Pacific (Mumbai) Tripti Tiwari (320503430430) Tripti_Tiwari5
Installing : libnftfilter_conntrack-1.0.8-2.amzn2023.0.2.x86_64 6/11
Installing : iptables-libs-1.8.8-3.amzn2023.0.2.x86_64 7/11
Installing : iptables-nft-1.8.8-3.amzn2023.0.2.x86_64 8/11
Running scriptlet: iptables-nft-1.8.8-3.amzn2023.0.2.x86_64 8/11
Installing : libgroup-3.0-1.amzn2023.0.1.x86_64 9/11
Running scriptlet: container-selinux-4:2.245.0-1.amzn2023.noarch 10/11
Installing : container-selinux-4:2.245.0-1.amzn2023.noarch 10/11
Running scriptlet: container-selinux-4:2.245.0-1.amzn2023.noarch 10/11
Running scriptlet: docker-25.0.14-1.amzn2023.0.6.x86_64 11/11
Installing : docker-25.0.14-1.amzn2023.0.6.x86_64 11/11
Running scriptlet: docker-25.0.14-1.amzn2023.0.6.x86_64 11/11
Created symlink /etc/systemd/system/sockets.target.wants/docker.socket → /usr/lib/systemd/system/docker.socket.

Running scriptlet: container-selinux-4:2.245.0-1.amzn2023.noarch 11/11
Running scriptlet: docker-25.0.14-1.amzn2023.0.6.x86_64 11/11
Verifying : container-selinux-4:2.245.0-1.amzn2023.noarch 1/11
Verifying : containerd-2.2.3-1.amzn2023.0.2.x86_64 2/11
Verifying : docker-25.0.14-1.amzn2023.0.6.x86_64 3/11
Verifying : iptables-libs-1.8.8-3.amzn2023.0.2.x86_64 4/11
Verifying : iptables-nft-1.8.8-3.amzn2023.0.2.x86_64 5/11
Verifying : libgroup-3.0-1.amzn2023.0.1.x86_64 6/11
Verifying : libnftfilter_conntrack-1.0.8-2.amzn2023.0.2.x86_64 7/11
Verifying : libnftnl-1.0.1-19.amzn2023.0.2.x86_64 8/11
Verifying : libnftnl-1.2.2-2.amzn2023.0.2.x86_64 9/11
Verifying : pigz-2.5-1.amzn2023.0.3.x86_64 10/11
Verifying : runc-1.3.4-5.amzn2023.0.2.x86_64 11/11

Installed:
container-selinux-4:2.245.0-1.amzn2023.noarch containerd-2.2.3-1.amzn2023.0.2.x86_64 docker-25.0.14-1.amzn2023.0.6.x86_64
iptables-libs-1.8.8-3.amzn2023.0.2.x86_64 iptables-nft-1.8.8-3.amzn2023.0.2.x86_64 libgroup-3.0-1.amzn2023.0.1.x86_64
libnftfilter_conntrack-1.0.8-2.amzn2023.0.2.x86_64 libnftnl-1.0.1-19.amzn2023.0.2.x86_64 libnftnl-1.2.2-2.amzn2023.0.2.x86_64
pigz-2.5-1.amzn2023.0.3.x86_64 runc-1.3.4-5.amzn2023.0.2.x86_64

Complete!
[ec2-user@ip-172-31-40-231 ~]$ sudo systemctl start docker
```

sudo systemctl start docker

```
Complete!
[ec2-user@ip-172-31-40-231 ~]$ sudo systemctl start docker
[ec2-user@ip-172-31-40-231 ~]$ sudo systemctl enable docker
Created symlink /etc/systemd/system/multi-user.target.wants/docker.service → /usr/lib/systemd/system/docker.service.
[ec2-user@ip-172-31-40-231 ~]$ sudo usermod -aG docker ec2-user
[ec2-user@ip-172-31-40-231 ~]$ docker --version
Docker version 25.0.14, build 0bab007
[ec2-user@ip-172-31-40-231 ~]$ docker run hello-world
docker: permission denied while trying to connect to the Docker daemon socket at unix:///var/run/docker.sock: Head "http://%2Fvar%2Frun%2Fdocker.sock/_ping":
dial unix /var/run/docker.sock: connect: permission denied.
See 'docker run --help'.
[ec2-user@ip-172-31-40-231 ~]$
```

i-0b9340441837baa29 (my new server)

PublicIPs: 13.201.26.131 PrivateIPs: 172.31.40.231

 CloudShell  Feedback  Console Mobile App

© 2026, Amazon Web Services, Inc. or its affiliates. [Privacy](#) [Terms](#) [Cookie preferences](#)

sudo systemctl enable docker

```
Complete!
[ec2-user@ip-172-31-40-231 ~]$ sudo systemctl start docker
[ec2-user@ip-172-31-40-231 ~]$ sudo systemctl enable docker
Created symlink /etc/systemd/system/multi-user.target.wants/docker.service → /usr/lib/systemd/system/docker.service.
[ec2-user@ip-172-31-40-231 ~]$ sudo usermod -aG docker ec2-user
[ec2-user@ip-172-31-40-231 ~]$ docker --version
Docker version 25.0.14, build 0bab007
[ec2-user@ip-172-31-40-231 ~]$ docker run hello-world
docker: permission denied while trying to connect to the Docker daemon socket at unix:///var/run/docker.sock: Head "http://%2Fvar%2Frun%2Fdocker.sock/_ping":
dial unix /var/run/docker.sock: connect: permission denied.
See 'docker run --help'.
[ec2-user@ip-172-31-40-231 ~]$
```

i-0b9340441837baa29 (my new server)

PublicIPs: 13.201.26.131 PrivateIPs: 172.31.40.231

 CloudShell  Feedback  Console Mobile App

© 2026, Amazon Web Services, Inc. or its affiliates. [Privacy](#) [Terms](#) [Cookie preferences](#)

STEP 3:- Grant Docker Permissions

sudo usermod -aG docker ec2-user

```
[ec2-user@ip-172-31-40-231 ~]$ sudo usermod -aG docker ec2-user
[ec2-user@ip-172-31-40-231 ~]$ docker --version
Docker version 25.0.14, build 0bab007
[ec2-user@ip-172-31-40-231 ~]$ docker run hello-world
docker: permission denied while trying to connect to the Docker daemon socket at unix:///var/run/docker.sock: Head "http://%2Fvar%2Frun%2Fdocker.sock/_ping":
dial unix /var/run/docker.sock: connect: permission denied.
See 'docker run --help'.
[ec2-user@ip-172-31-40-231 ~]$ sudo docker run hello-world
Unable to find image 'hello-world:latest' locally
latest: Pulling from library/hello-world
4f55086f7dd0: Pull complete
Digest: sha256:0e760fdbc48ba8041e7c6db999bb40bfca508b4be580ac75d32c4e29d202ce1
Status: Downloaded newer image for hello-world:latest

Hello from Docker!
This message shows that your installation appears to be working correctly.

To generate this message, Docker took the following steps:
1. The Docker client contacted the Docker daemon.
2. The Docker daemon pulled the "hello-world" image from the Docker Hub.
   (amd64)
3. The Docker daemon created a new container from that image which runs the
   executable that produces the output you are currently reading.
4. The Docker daemon streamed that output to the Docker client, which sent it
   to your terminal.

To try something more ambitious, you can run an Ubuntu container with:
$ docker run -it ubuntu bash

Share images, automate workflows, and more with a free Docker ID:
https://hub.docker.com/

For more examples and ideas, visit:
https://docs.docker.com/get-started/

[ec2-user@ip-172-31-40-231 ~]$
```

i-0b9340441837baa29 (my new server)

PublicIPs: 13.201.26.131 PrivateIPs: 172.31.40.231

CloudShell Feedback Console Mobile App

© 2026, Amazon Web Services, Inc. or its affiliates. Privacy Terms Cookie preferences

STEP 4:- Verify Docker

docker --version

docker run hello-world

```
[ec2-user@ip-172-31-40-231 ~]$ sudo usermod -aG docker ec2-user
[ec2-user@ip-172-31-40-231 ~]$ docker --version
Docker version 25.0.14, build 0bab007
[ec2-user@ip-172-31-40-231 ~]$ docker run hello-world
docker: permission denied while trying to connect to the Docker daemon socket at unix:///var/run/docker.sock: Head "http://%2Fvar%2Frun%2Fdocker.sock/_ping":
dial unix /var/run/docker.sock: connect: permission denied.
See 'docker run --help'.
[ec2-user@ip-172-31-40-231 ~]$ sudo docker run hello-world
Unable to find image 'hello-world:latest' locally
latest: Pulling from library/hello-world
4f55086f7dd0: Pull complete
Digest: sha256:0e760fdbc48ba8041e7c6db999bb40bfca508b4be580ac75d32c4e29d202ce1
Status: Downloaded newer image for hello-world:latest

Hello from Docker!
This message shows that your installation appears to be working correctly.

To generate this message, Docker took the following steps:
1. The Docker client contacted the Docker daemon.
2. The Docker daemon pulled the "hello-world" image from the Docker Hub.
   (amd64)
3. The Docker daemon created a new container from that image which runs the
   executable that produces the output you are currently reading.
4. The Docker daemon streamed that output to the Docker client, which sent it
   to your terminal.

To try something more ambitious, you can run an Ubuntu container with:
$ docker run -it ubuntu bash

Share images, automate workflows, and more with a free Docker ID:
https://hub.docker.com/

For more examples and ideas, visit:
https://docs.docker.com/get-started/

[ec2-user@ip-172-31-40-231 ~]$
```

i-Ob9340441837baa29 (my new server)

PublicIPs: 13.201.26.131 PrivateIPs: 172.31.40.231

CloudShell Feedback Console Mobile App

© 2026, Amazon Web Services, Inc. or its affiliates. Privacy Terms Cookie preferences

If Docker is installed correctly, you'll see the Docker Hello World message.

THEN, Reconnect to the instance and verify:
groups

EC2 > Instances > i-0b9340441837baa29

EC2

- Dashboard
- AWS Global View
- Events
- ▼ Instances
 - Instances
 - Instance Types
 - Launch Templates
 - Spot Requests
 - Savings Plans
 - Reserved Instances
 - Dedicated Hosts
 - Capacity Reservations
 - Capacity Manager
- ▼ Images
 - AMIs
 - AMI Catalog
- ▼ Elastic Block Store
 - Volumes
 - Snapshots
 - Lifecycle Manager

Instance summary for i-0b9340441837baa29 (my new server)

Updated less than a minute ago

Instance ID i-0b9340441837baa29	Public IPv4 address 13.201.26.131 open address	Private IPv4 addresses 172.31.40.231
IPv6 address -	Instance state Running	Public DNS ec2-13-201-26-131.ap-south-1.compute.amazonaws.com open address
Hostname type IP name: ip-172-31-40-231.ap-south-1.compute.internal	Private IP DNS name (IPv4 only) ip-172-31-40-231.ap-south-1.compute.internal	Elastic IP addresses -
Answer private resource DNS name IPv4 (A)	Instance type t3.micro	AWS Compute Optimizer finding Opt-in to AWS Compute Optimizer for recommendations. Learn more
Auto-assigned IP address 13.201.26.131 [Public IP]	VPC ID vpc-0dcf2bc6a564abce9	Auto Scaling Group name -
IAM role -	Subnet ID subnet-0a7b18c37ddae3670	Managed false
IMDSv2 Required	Instance ARN arn:aws:ec2:ap-south-1:320503430430:instance/i-0b9340441837baa29	

© 2026, Amazon Web Services, Inc. or its affiliates. Privacy Terms Cookie preferences

You have to modify the IAM role from your instances

aws

Search [Option+S]

Asia Pacific (Mumbai) Tripti Tiwari (320503430430) Tripti_Tiwari5

EC2 > Instances > i-0b9340441837baa29 > Modify IAM role

Modify IAM role

Attach an IAM role to your instance.

Instance ID
i-0b9340441837baa29 (my new server)

IAM role
Select an IAM role to attach to your instance or create a new role if you haven't created any. The role you select replaces any roles that are currently attached to your instance.

ec2-service

Create new IAM role

Cancel Update IAM role

CloudShell Feedback Console Mobile App

© 2026, Amazon Web Services, Inc. or its affiliates. Privacy Terms Cookie preferences

Then run:

docker run hello-world

```
aws [Option+S] Asia Pacific (Mumbai) Tripti Tiwari (320503430410) Tripti_Tiwari5

[ec2-user@ip-172-31-40-231 ~]$ groups
ec2-user adm wheel systemd-journal
[ec2-user@ip-172-31-40-231 ~]$ newgrp docker
[ec2-user@ip-172-31-40-231 ~]$ docker run hello-world

Hello from Docker!
This message shows that your installation appears to be working correctly.

To generate this message, Docker took the following steps:
1. The Docker client contacted the Docker daemon.
2. The Docker daemon pulled the "hello-world" image from the Docker Hub.
   (amd64)
3. The Docker daemon created a new container from that image which runs the
   executable that produces the output you are currently reading.
4. The Docker daemon streamed that output to the Docker client, which sent it
   to your terminal.

To try something more ambitious, you can run an Ubuntu container with:
$ docker run -it ubuntu bash

Share images, automate workflows, and more with a free Docker ID:
https://hub.docker.com/

For more examples and ideas, visit:
https://docs.docker.com/get-started/

[ec2-user@ip-172-31-40-231 ~]$ groups
docker adm wheel systemd-journal ec2-user
[ec2-user@ip-172-31-40-231 ~]$ ec2-user wheel docker
bash: ec2-user: command not found
[ec2-user@ip-172-31-40-231 ~]$ docker pull nginx
Using default tag: latest
latest: Pulling from library/nginx
5b4d6ff92fc4: Pull complete
830625e1ac85: Pull complete
5431d0092ffd: Pull complete
b4a248c845e5: Pull complete
7f8b1a2b17d8: Pull complete
45381ecb0e2f: Pull complete
13fd728be9eb: Pull complete
Digest: sha256:5aca99593157f4ae539a5dec1092a0ad8762f8e2eb1789085a13a0f5622369f6
Status: Downloaded newer image for nginx:latest
docker.io/library/nginx:latest
[ec2-user@ip-172-31-40-231 ~]$ docker images
```

i-0b9340441837baa29 (my new server)
PublicIPs: 13.201.26.131 PrivateIPs: 172.31.40.231

CloudShell Feedback Console Mobile App © 2026, Amazon Web Services, Inc. or its affiliates. Privacy Terms Cookie preferences

STEP 5:- Important Docker Commands

docker pull nginx

```
aws [Option+S] Asia Pacific (Mumbai) Tripti Tiwari (320503430430) Tripti_Tiwari5

[ec2-user@ip-172-31-40-231 ~]$ ec2-user wheel docker
bash: ec2-user: command not found
[ec2-user@ip-172-31-40-231 ~]$ docker pull nginx
Using default tag: latest
latest: Pulling from library/nginx
5b4d6ff92fc4: Pull complete
830625e1ac85: Pull complete
5431d0092ffd: Pull complete
b4a248c845e5: Pull complete
7f8b1a2b17d8: Pull complete
45381ecb0e2f: Pull complete
13fd728be9eb: Pull complete
Digest: sha256:5aca99593157f4ae539a5dec1092a0ad8762f8e2eb1789085a13a0f5622369f6
Status: Downloaded newer image for nginx:latest
docker.io/library/nginx:latest
[ec2-user@ip-172-31-40-231 ~]$ docker images
```

docker images


```

5b4d6ff92fc4: Pull complete
830625e1ac85: Pull complete
5431d0092ffd: Pull complete
b4a248c845e5: Pull complete
7f8b1a2b1d8: Pull complete
45381ecb0e2f: Pull complete
13fd728be9eb: Pull complete
Digest: sha256:5aca99593157f4ae539a5dec1092a0ad8762f8e2eb1789085a13a0f5622369f6
Status: Downloaded newer image for nginx:latest
docker.io/library/nginx:latest
[ec2-user@ip-172-31-40-231 ~]$ docker images
REPOSITORY      TAG        IMAGE ID       CREATED        SIZE
nginx           latest     7aaca76c508f   13 days ago    161MB
hello-world     latest     e2ac70e7319a   2 months ago   10.1kB
[ec2-user@ip-172-31-40-231 ~]$

```

docker ps docker ps -a

```

13fd728be9eb: Pull complete
Digest: sha256:5aca99593157f4ae539a5dec1092a0ad8762f8e2eb1789085a13a0f5622369f6
Status: Downloaded newer image for nginx:latest
docker.io/library/nginx:latest
[ec2-user@ip-172-31-40-231 ~]$ docker images
REPOSITORY      TAG        IMAGE ID       CREATED        SIZE
nginx           latest     7aaca76c508f   13 days ago    161MB
hello-world     latest     e2ac70e7319a   2 months ago   10.1kB
[ec2-user@ip-172-31-40-231 ~]$ docker ps
CONTAINER ID   IMAGE      COMMAND                  CREATED    STATUS    PORTS    NAMES
[ec2-user@ip-172-31-40-231 ~]$ docker ps -a
CONTAINER ID   IMAGE      COMMAND                  CREATED    STATUS    PORTS    NAMES
8555d617b625   hello-world "/hello"                3 minutes ago    Exited (0) 3 minutes ago    festive_blackwell
c7bfcaa2bf7d   hello-world "/hello"                6 minutes ago    Exited (0) 6 minutes ago    tender_rhodes
[ec2-user@ip-172-31-40-231 ~]$

```

i-0b9340441837baa29 (my new server)

docker stop <container-id>

```

45381ecb0e2f: Pull complete
13fd728be9eb: Pull complete
Digest: sha256:5aca99593157f4ae539a5dec1092a0ad8762f8e2eb1789085a13a0f5622369f6
Status: Downloaded newer image for nginx:latest
docker.io/library/nginx:latest
[ec2-user@ip-172-31-40-231 ~]$ docker images
REPOSITORY      TAG        IMAGE ID       CREATED        SIZE
nginx           latest     7aaca76c508f   13 days ago    161MB
hello-world     latest     e2ac70e7319a   2 months ago   10.1kB
[ec2-user@ip-172-31-40-231 ~]$ docker ps
CONTAINER ID   IMAGE      COMMAND                  CREATED    STATUS    PORTS    NAMES
[ec2-user@ip-172-31-40-231 ~]$ docker ps -a
CONTAINER ID   IMAGE      COMMAND                  CREATED    STATUS    PORTS    NAMES
8555d617b625   hello-world "/hello"                3 minutes ago    Exited (0) 3 minutes ago    festive_blackwell
c7bfcaa2bf7d   hello-world "/hello"                6 minutes ago    Exited (0) 6 minutes ago    tender_rhodes
[ec2-user@ip-172-31-40-231 ~]$ docker stop 8555d617b625
8555d617b625
[ec2-user@ip-172-31-40-231 ~]$ docker rm 8555d617b625
8555d617b625
[ec2-user@ip-172-31-40-231 ~]$ docker rmi
[ec2-user@ip-172-31-40-231 ~]$

```

i-0b9340441837baa29 (my new server)

docker rm <container-id>

```

8555d617b625   hello-world "/hello"                3 minutes ago    Exited (0) 3 minutes ago    festive_blackwell
c7bfcaa2bf7d   hello-world "/hello"                6 minutes ago    Exited (0) 6 minutes ago    tender_rhodes
[ec2-user@ip-172-31-40-231 ~]$ docker stop 8555d617b625
8555d617b625
[ec2-user@ip-172-31-40-231 ~]$ docker rm 8555d617b625
8555d617b625

```

docker rmi <image-id>

```

45381ecb0e2f: Pull complete
13fd728be9eb: Pull complete
Digest: sha256:5aca99593157f4ae539a5dec1092a0ad8762f8e2eb1789085a13a0f5622369f6
Status: Downloaded newer image for nginx:latest
docker.io/library/nginx:latest
[ec2-user@ip-172-31-40-231 ~]$ docker images
REPOSITORY    TAG       IMAGE ID       CREATED        SIZE
nginx         latest   7aaca76c508f   13 days ago    161MB
hello-world   latest   e2ac70e7319a   2 months ago   10.1kB
[ec2-user@ip-172-31-40-231 ~]$ docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS   NAMES
[ec2-user@ip-172-31-40-231 ~]$ docker ps -a
CONTAINER ID   IMAGE     COMMAND   CREATED        STATUS      PORTS   NAMES
8555d617b625   hello-world   "/hello"   3 minutes ago   Exited (0) 3 minutes ago
c7bfcaa2bf7d   hello-world   "/hello"   6 minutes ago   Exited (0) 6 minutes ago
[ec2-user@ip-172-31-40-231 ~]$ docker stop 8555d617b625
8555d617b625
[ec2-user@ip-172-31-40-231 ~]$ docker rm 8555d617b625
8555d617b625
[ec2-user@ip-172-31-40-231 ~]$ docker rmi

```

STEP 6:- Portfolio Website using Docker

Create index.html

```

<!DOCTYPE html>

<html>

<head> <title>Student Portfolio</title>

</head>

<body>

<h1>Welcome to My Portfolio</h1>

<p>Name: Student Name</p>

<p>Skills: AWS, Python, Docker</p>

</body>

</html>

```

```
Error response from daemon: conflict: unable to remove repository reference "hello-world" (must force) - container c7bfcaa2bf7d is using its referenced image e2ac70e7319a
[ec2-user@ip-172-31-40-231 ~]$ mkdir student-portfolio
[ec2-user@ip-172-31-40-231 ~]$ cd student-portfolio
[ec2-user@ip-172-31-40-231 student-portfolio]$ nano index.html
[ec2-user@ip-172-31-40-231 student-portfolio]$ cat index.html
<!DOCTYPE html>
<html>
<head>
<title>Student Portfolio</title>
</head>
<body>
<h1>Welcome to My Portfolio</h1>
<p>Name: Student Name</p>
<p>Skills: AWS, Python, Docker</p>
</body>
</html>
[ec2-user@ip-172-31-40-231 student-portfolio]$
```

i-0b9340441837baa29 (my new server)

PublicIPs: 13.201.26.131 PrivateIPs: 172.31.40.231

CloudShell Feedback Console Mobile App

© 2026, Amazon Web Services, Inc. or its affiliates. Privacy Terms Cookie preferences

STEP 7:- Create Dockerfile

FROM nginx:latest

COPY . /usr/share/nginx/html

```
[ec2-user@ip-172-31-40-231 student-portfolio]$ cat Dockerfile
FROM nginx:latest
COPY . /usr/share/nginx/html
[ec2-user@ip-172-31-40-231 student-portfolio]$ sudo docker build -t student-portfolio .
[+] Building 0.4s (7/7) FINISHED
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 143B
=> [internal] load metadata for docker.io/library/nginx:latest
=> [internal] load .dockerignore
=> => transferring context: 2B
=> [internal] load build context
=> => transferring context: 431B
=> [1/2] FROM docker.io/library/nginx:latest
=> [2/2] COPY . /usr/share/nginx/html
=> exporting to image
=> => exporting layers
=> => writing image sha256:1a408b50e8f7e7f5ac3e78b3d798dabfba968eb559779d2d821ald67c8b678ab
=> => naming to docker.io/library/student-portfolio
[ec2-user@ip-172-31-40-231 student-portfolio]$
```

i-0b9340441837baa29 (my new server)

PublicIPs: 13.201.26.131 PrivateIPs: 172.31.40.231

CloudShell Feedback Console Mobile App

© 2026, Amazon Web Services, Inc. or its affiliates. Privacy Terms Cookie preferences

STEP 8:- Build Docker Image

docker build -t student-portfolio .

```

[ec2-user@ip-172-31-40-231 student-portfolio]$ sudo docker build -t student-portfolio .
[+] Building 0.4s (7/7) FINISHED
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 143B
=> [internal] load metadata for docker.io/library/nginx:latest
=> [internal] load .dockerignore
=> => transferring context: 2B
=> [internal] load build context
=> => transferring context: 431B
=> [1/2] FROM docker.io/library/nginx:latest
=> [2/2] COPY . /usr/share/nginx/html
=> exporting to image
=> => exporting layers
=> => writing image sha256:1a408b50e8f7e7f5ac3e78b3d798dabfba968eb559779d2d821ald67c8b678ab
=> => naming to docker.io/library/student-portfolio
[ec2-user@ip-172-31-40-231 student-portfolio]$ sudo docker run -d -p 80:80 student-portfolio
```

STEP 9:- Run Docker Container

docker run -d -p 80:80 student-portfolio

```

=> => naming to docker.io/library/student-portfolio
[ec2-user@ip-172-31-40-231 student-portfolio]$ sudo docker run -d -p 80:80 student-portfolio
01a14a57af526408fb6919809d5fc9c27d026de58636b473d0a2ffbc131a507
[ec2-user@ip-172-31-40-231 student-portfolio]$ sudo docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS                               NAMES
01a14a57af52   student-portfolio "/docker-entrypoint..." 26 seconds ago Up 25 seconds 0.0.0.0:80->80/tcp, :::80->80/tcp   inspiring_robinson
[ec2-user@ip-172-31-40-231 student-portfolio]$ sudo docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS                               NAMES
01a14a57af52   student-portfolio "/docker-entrypoint..." 9 minutes ago   Up 9 minutes   0.0.0.0:80->80/tcp, :::80->80/tcp   inspiring_robinson
[ec2-user@ip-172-31-40-231 student-portfolio]$ curl localhost
```

STEP 10:- Verify Website

Open browser and visit <http://localhost> on Windows or <http://> on AWS.

```

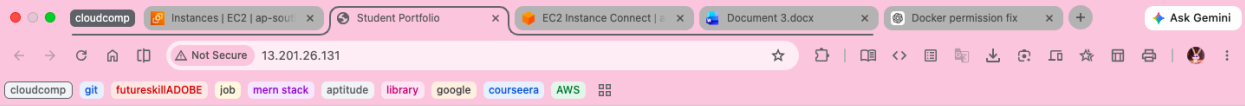
[ec2-user@ip-172-31-40-231 student-portfolio]$ curl localhost
<!DOCTYPE html>
<html>
<head>
<title>Student Portfolio</title>
</head>
<body>
<h1>Welcome to My Portfolio</h1>
<p>Name: Student Name</p>
<p>Skills: AWS, Python, Docker</p>
</body>
</html>
[ec2-user@ip-172-31-40-231 student-portfolio]$ curl http://localhost
<!DOCTYPE html>
<html>
<head>
<title>Student Portfolio</title>
</head>
<body>
<h1>Welcome to My Portfolio</h1>
<p>Name: Student Name</p>
<p>Skills: AWS, Python, Docker</p>
</body>
</html>
[ec2-user@ip-172-31-40-231 student-portfolio]$ sudo docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS                               NAMES
01a14a57af52   student-portfolio "/docker-entrypoint..." 12 minutes ago Up 12 minutes 0.0.0.0:80->80/tcp, :::80->80/tcp   inspiring_robinson
[ec2-user@ip-172-31-40-231 student-portfolio]$
```

i-0b9340441837baa29 (my new server)

PublicIPs: 13.201.26.131 PrivateIPs: 172.31.40.231

CloudShell Feedback Console Mobile App

© 2026, Amazon Web Services, Inc. or its affiliates. Privacy Terms Cookie preferences



Welcome to My Portfolio

Name: Student Name

Skills: AWS, Python, Docker